Suleyman Inaltekin

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Statistics Ph.D. student at Rice interested in physics-based AI: energy-based and probabilistic models, sampling as computation, and the hardware–software co-design that makes AI orders of magnitude more efficient. Builds models from scratch in PyTorch and NumPy; trained in causal inference via political science. Seeking a Summer 2027 research internship in physics-inspired or unconventional AI.

Education

Rice University — Houston, TX
Ph.D. in Statistics | *Advisor: Prof. Meng Li* — *Aug 2024 – Present (expected May 2029)*
- Transferred from the Political Science Ph.D. program (2023–24) to focus on statistical machine learning.
- Coursework: Statistical Learning, Deep Learning, Bayesian Inference, Causal Inference, Probability

University of Vienna — Vienna, Austria
B.Sc. Mathematics and B.A. Political Science | *GPA: 3.7/4.0* — *2019 – 2023*

Memorial International School of Tirana — Tirana, Albania
Cambridge International A Levels: Mathematics (A), History (A*), Psychology (A*)* — *2019*

Research Experience

Graduate Researcher, Department of Statistics, Rice University — 2024 – Present
Project title, e.g., Langevin and Oscillator-Based Samplers for Energy-Based Models — *Advisor: Prof. Meng Li*
- Developing [method] for [problem]; [result, e.g., 5× faster than MCMC on 1M rows at equal accuracy].
- Released [method] as an open-source R/Python package; [presented at JSM 2026 / paper in prep].

Projects

PunchTracker: an AI Boxing Coach from a Phone Video | code — Python, MediaPipe, OpenCV, SciPy
- Built a computer-vision pipeline that tracks 33 body keypoints per frame and detects, counts, and classifies punches (jab, cross, hook, uppercut) from a single smartphone video, with no wearable sensors.
- Turned wrist-extension trajectories into a time series and detected punches with velocity-thresholded peak finding; [X]% count accuracy against hand-labeled sparring rounds.

Swipe or Stay: Predicting Attention Retention in Short Videos | code — PyTorch, pandas
- Trained a neural embedding model on 4.7M user–video interactions (KuaiRec) to predict watch time, a core engagement signal in short-video feed ranking (TikTok, Reels, Spotlight).
- Diagnosed and corrected duration bias (longer videos mechanically get lower watch ratios) with duration-conditional labels; [X]% lower error than matrix-factorization and popularity baselines.

Autograd & Neural Networks from Scratch | code — Python, NumPy
- Built a micrograd-style autograd engine and MLP in pure NumPy; verified gradients against finite differences and reached [X]% test accuracy on MNIST.

Teaching

Teaching Assistant, Rice University | **STAT 3XX: Course Name** — Fall 2024 – Present

Skills

Programming: Python (PyTorch, NumPy, pandas, scikit-learn, OpenCV), R (tidyverse, Stan), SQL, Git, LaTeX
Methods: causal inference (DiD, IV, RDD, matching), A/B testing & experimental design, Bayesian modeling, MCMC & Langevin sampling, energy-based models, deep learning, computer vision
Languages: Albanian (native), English (fluent), German (professional)
Interests: boxing, chess, sewing, card magic, neuromorphic computing